

AFC Title-Filtered Patent Report

Expanded explanation: what each patent/application did

Scope: title contains AFC, Automatic Frequency Coordination, Automated Frequency Coordination, or Standard Power. Prepared July 13, 2026.

Purpose of this version

This update adds a plain-English explanation of what each patent/application actually did: the problem it solved, the AFC mechanism it introduced, and why it matters. The list remains title-filtered as requested.

Important limitation: this is a public-source technical landscape, not a legal freedom-to-operate opinion. For legal use, pull full claims, prosecution history, assignments, and active legal status directly from patent-office databases.

Executive Summary Table

No.	Patent / family	Title	Assignee	What it did
1	US20200236558A1 / US11510065B2 / WO2020154406A1	Automatic frequency coordination (AFC) for unlicensed spectrum	Qualcomm Inc.	It created a foundational AFC request-response model for unlicensed/shared spectrum. A device or AP determines that it needs AFC authorization, requests spectrum information, receives channel/power constraints, and modifies its wireless operation accordingly. This is one of the basic patterns behind standard-power 6 GHz operation: do not transmit at higher power until AFC has returned allowed operating parameters.
2	US20200260289A1 / US11252574B2	Automated frequency coordination for shared spectrum wireless systems	Federated Wireless Inc.	It built the AFC service-provider side. The system gathers incumbent/high-priority user parameters, calculates interference-to-noise contours, stores those contours in a database, receives RLAN/channel availability requests, and returns frequency availability. It made AFC scalable by using stored contour data instead of a purely real-time interference calculation for every request.
3	US11234135	Automated frequency coordination for shared spectrum wireless systems	Federated Wireless Inc.	It appears to cover a related Federated Wireless shared-spectrum AFC system, again centered on high-priority user protection, contour calculation, and spectrum availability responses. It should be treated as a likely related or parallel family to the Federated AFC database approach until full family records are pulled.
4	US20220224496A1 / US12069000B2	Systems and methods for automatic frequency coordination	Outdoor Wireless Networks LLC / CommScope history	It focused on exclusion zones and incumbent receiver protection. The method uses receiver/channel/antenna/radial information to decide which channels or power levels are unavailable near protected fixed-service links. This turns regulatory protection rules into specific geographic/channel restrictions for AFC operation.
5	WO2022150475A1	Systems and methods for automatic frequency coordination	CommScope Technologies LLC	It is the international/PCT counterpart for the exclusion-zone AFC concept. It extends the same general invention into international patent-family strategy and should be used to identify national-phase rights and claim variations outside the US.
6	US20230022388A1 / US11589265B2	Iterative automated frequency coordination	Cisco Technology Inc.	It improved AFC results by repeatedly running AFC with different location-error margins. The system compares outcomes such as allowed-channel count and selects a better margin. It addresses the practical issue that overly conservative location uncertainty can unnecessarily reduce available channels, while too little uncertainty may fail incumbent protection.
7	WO2023009973A1 / EP4378192A1 / CN116438742B	Iterative automated frequency coordination	Cisco Technology Inc.	It protects Cisco iterative AFC outside the US. These family members matter because enterprise WLAN controllers and AP deployments are global, and location-uncertainty optimization can affect AFC outcomes in multiple regulatory regimes.

No.	Patent / family	Title	Assignee	What it did
8	US20210120556A1 / US12317239B2	Apparatus, system and method of communication according to an automated frequency coordination (AFC) allocation scheme	Intel Corp.	It mapped AFC allocation into real wireless communication behavior. Instead of only calculating available spectrum, it describes apparatus/system procedures for communicating according to an AFC allocation scheme, including station/AP behavior and resource use. This makes it relevant for chipset and device implementation.
9	US20220329979A1	Automated frequency coordination and device location awareness	Arris Enterprises LLC	It tied AFC authorization to device location awareness. The device determines or validates its physical location and uses that information in AFC interaction. This solves a basic AFC requirement: standard-power operation depends on the AFC system knowing where the AP actually is.
10	US20240098615A1 / US12375995B2	Automated frequency coordination for high speed mobility	Qualcomm Incorporated	It adapted AFC for moving/high-speed devices. A device can send AFC requests along a path, use AFC responses to set power, and reduce repeated requests on later trips along the same route. This targets trains, vehicles, robots, or other recurring mobility paths where static AFC assumptions are insufficient.
11	WO2024058874A1 / EP4588288A1 / CN119836828A	Automated frequency coordination for high speed mobility	Qualcomm Incorporated	It extends the Qualcomm high-speed mobility AFC invention internationally. The family is important for global connected-transport and industrial-mobility products where the same moving AP/client concepts may be deployed in multiple countries.
12	US20240171988A1 / US12563405	Automated frequency coordination (AFC) negotiated band access	Cisco Technology Inc.	It changed AFC from pure immediate availability lookup into negotiated future access. A WLAN provider requests spectrum for a future time window; the AFC/coordination system queries an incumbent; if granted, permission is returned to the requester. This supports scheduled, reservation-based, or incumbent-approved spectrum use.
13	US20240334416A1 / US12439395B2	Opportunistic standard power grant by automated frequency coordination	Cisco Technology Inc.	It lets an AP start in low-power mode and opportunistically move to standard-power mode when conditions allow. The AP uses a STA/client location token, sends location information to AFC, receives permission, and switches operating mode. This can increase range/performance while staying AFC-compliant.
14	US20260136350A1	Opportunistic standard power grant by automated frequency coordination	Cisco Technology Inc.	It is a continuation of the opportunistic standard-power grant family. Continuations often pursue additional claim scope around implementation details, such as token handling, STA location use, AP mode switching, or AFC instruction processing.
15	WO2024153114	Automated frequency coordination in wireless communication networks	MediaTek-related PCT filing context	It moved AFC into a multi-AP network-controller architecture. APs or the controller obtain spectrum availability information based on location, and the controller configures multiple APs accordingly. This is relevant to enterprise Wi-Fi and managed networks where AFC should be coordinated across many APs.
16	US20240388410A1	Automated Frequency Coordination (AFC) system and method for controlling frequency sharing and/or interference in a communications system	Charter Communications Operating LLC	It created a precomputed threshold/interference decision approach. The system builds threshold-to-interference tables from transmitter/receiver characteristics before requests arrive, then uses those tables to decide whether a transmitter would cause unacceptable interference. This speeds AFC decisions.
17	US20250088294A1	Monitoring of automated frequency coordination network to detect and mitigate successive interference	Not confirmed from snippet	It added ongoing monitoring after AFC grants. The system watches interference indicators and AP KPI data, identifies likely interfering APs, and takes mitigation actions. This is useful because interference problems can appear after initial AFC authorization.

No.	Patent / family	Title	Assignee	What it did
18	US20240349173A1 / US12598546B2	Successive interference mitigation in automated frequency coordination (AFC)	Cisco Technology Inc.	It identifies and mitigates aggregate or successive interference from APs. It receives AP channel-use messages, detects when an incumbent fixed-service channel exceeds an I/N threshold, identifies AP contributors, and takes action to reduce the interference.
19	US20250039912A1 / US12550172B2	Destination-based automated frequency coordination	Cisco Technology Inc.	It bases AFC frequency lookup on the destination or mobile client location rather than only the AP location. The system determines the mobile client location, asks AFC for availability at that location, selects a frequency, and causes the client to use it.
20	US20250097945A1	Automatic frequency coordination with residential gateway access points	Charter Communications Operating LLC	It embeds AFC/control logic into residential or business gateway APs. The gateway applies local channel exclusion and self-organizes around unavailable unlicensed channels. This targets cable/MSO gateway deployments where AFC needs to work at home/business scale.
21	US20250247817A1	Optimizing automated frequency coordination for fixed path uses	Cisco Technology Inc.	It optimizes AFC for known fixed routes. The system models path segments/polygons, predicts channel/power availability along the route, and selects channels proactively based on upcoming conditions. It targets rail, guided vehicles, warehouses, mines, and campus transport.
22	US20250274874A1	Automatic frequency coordination	Qualcomm Inc.	It appears to add context-aware AFC using position/height and radio-access information. The system uses device context to constrain transmit power/channel selection under AFC. This can benefit chipset implementations where height and RAT context affect allowed operation.
23	US20260040194A1	Automated Frequency Coordination Registration	Cisco Technology Inc.	It focuses on registering APs or network devices with an AFC system. Device identity, location, and network information are used to onboard equipment efficiently. This is significant for large AP fleets where AFC registration must scale.
24	US20260046832A1	Request of automated frequency coordination response	Hewlett Packard Enterprise Development LP	It improves how an AP requests an AFC response using location samples, GPS/ranging data, or server-side processing. The goal is to produce a faster or more accurate AFC response based on stronger location evidence.
25	US20260082363A1	Automated frequency coordination (AFC) and geolocation enhancements	Qualcomm Inc.	It improves AFC by querying multiple height/geolocation services and choosing the smallest vertical uncertainty. Since AFC decisions can depend on AP height and uncertainty, better vertical confidence may increase usable channels or power while maintaining protection.
26	WO2025250411	Automated frequency coordination authentication	Not confirmed from snippet	It addresses trust and enforcement around AFC response handling. The device receives available spectrum/power constraints, determines when to update its channel/power based on wireless interface state, and applies the updated operating parameters.

Detailed Records - What Each Patent Did

1. Automatic frequency coordination (AFC) for unlicensed spectrum

Publication / family	US20200236558A1 / US11510065B2 / WO2020154406A1
Assignee	Qualcomm Inc.
Inventors	Aleksandar Damnjanovic; Yongbin Wei
Status	US published; US granted; WO counterpart
Priority	2019-01-22
Filing	2020-01-21
Publication / grant	2020-07-23; grant 2022-11-22
Technical category	Device-side AFC authorization; unlicensed 6 GHz access; channel/power response enforcement.

What it did: It created a foundational AFC request-response model for unlicensed/shared spectrum. A device or AP determines that it needs AFC authorization, requests spectrum information, receives channel/power constraints, and modifies its wireless operation accordingly. This is one of the basic patterns behind standard-power 6 GHz operation: do not transmit at higher power until AFC has returned allowed operating parameters.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

2. Automated frequency coordination for shared spectrum wireless systems

Publication / family	US20200260289A1 / US11252574B2
Assignee	Federated Wireless Inc.
Inventors	Samuel Jay Macmullan; Michael Oliver Ghorbanzadeh; James Ni; Kurt Schaubach
Status	Published; granted counterpart
Priority	2020-02-12
Filing	2020-02-12
Publication / grant	2020-08-13; grant 2022-02-15 reported
Technical category	AFC database; incumbent protection; I/N contours; channel availability responses.

What it did: It built the AFC service-provider side. The system gathers incumbent/high-priority user parameters, calculates interference-to-noise contours, stores those contours in a database, receives RLAN/channel availability requests, and returns frequency availability. It made AFC scalable by using stored contour data instead of a purely real-time interference calculation for every request.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

3. Automated frequency coordination for shared spectrum wireless systems

Publication / family	US11234135
Assignee	Federated Wireless Inc.
Inventors	Not confirmed in snippet
Status	Granted US patent
Priority	2019-05-06 provisional noted
Filing	2020 filing reported
Publication / grant	Issued 2022-01-25
Technical category	Shared-spectrum AFC; high-priority user modeling; contour-based decisions.

What it did: It appears to cover a related Federated Wireless shared-spectrum AFC system, again centered on high-priority user protection, contour calculation, and spectrum availability responses. It should be treated as a likely related or parallel family to the Federated AFC database approach until full family records are pulled.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

4. Systems and methods for automatic frequency coordination

Publication / family	US20220224496A1 / US12069000B2
Assignee	Outdoor Wireless Networks LLC / CommScope history
Inventors	Suryanarayana A. Kalenahalli; Ariful Hannan
Status	Published; granted active
Priority	2021-01-08
Filing	2022-01-06
Publication / grant	2022-07-14; grant 2024-08-20
Technical category	Exclusion-zone AFC; fixed-service protection; channel/power denial zones.

What it did: It focused on exclusion zones and incumbent receiver protection. The method uses receiver/channel/antenna/radial information to decide which channels or power levels are unavailable near protected fixed-service links. This turns regulatory protection rules into specific geographic/channel restrictions for AFC operation.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

5. Systems and methods for automatic frequency coordination

Publication / family	WO2022150475A1
Assignee	CommScope Technologies LLC
Inventors	Suryanarayana Kalenahalli; Ariful Hannan
Status	PCT publication
Priority	2021-01-08
Filing	2022-01-06
Publication / grant	2022-07-14
Technical category	International family for exclusion-zone AFC.

What it did: It is the international/PCT counterpart for the exclusion-zone AFC concept. It extends the same general invention into international patent-family strategy and should be used to identify national-phase rights and claim variations outside the US.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

6. Iterative automated frequency coordination

Publication / family	US20230022388A1 / US11589265B2
Assignee	Cisco Technology Inc.
Inventors	Vishal S. Desai; Jerome Henry
Status	Published; granted active
Priority	2021-07-26
Filing	2021-07-26
Publication / grant	2023-01-26; grant 2023-02-21
Technical category	Iterative AFC; location uncertainty tuning; AP/controller optimization.

What it did: It improved AFC results by repeatedly running AFC with different location-error margins. The system compares outcomes such as allowed-channel count and selects a better margin. It addresses the practical issue that overly conservative location uncertainty can unnecessarily reduce available channels, while too little uncertainty may fail incumbent protection.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

7. Iterative automated frequency coordination

Publication / family	WO2023009973A1 / EP4378192A1 / CN116438742B
Assignee	Cisco Technology Inc.
Inventors	Vishal S. Desai; Jerome Henry
Status	PCT/EP/CN family members
Priority	2021-07-26
Filing	PCT 2022-07-22
Publication / grant	WO 2023-02-02
Technical category	International family for iterative AFC.

What it did: It protects Cisco iterative AFC outside the US. These family members matter because enterprise WLAN controllers and AP deployments are global, and location-uncertainty optimization can affect AFC outcomes in multiple regulatory regimes.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

8. Apparatus, system and method of communication according to an automated frequency coordination (AFC) allocation scheme

Publication / family	US20210120556A1 / US12317239B2
Assignee	Intel Corp.
Inventors	Jonathan Segev; Qinghua Li; Hassan Yaghoobi; Emily H. Qi; Feng Jiang
Status	Published; granted active
Priority	2020-12-24
Filing	2020-12-24
Publication / grant	2021-04-22; grant 2025-05-27
Technical category	AFC allocation scheme; station/AP behavior; chipset implementation.

What it did: It mapped AFC allocation into real wireless communication behavior. Instead of only calculating available spectrum, it describes apparatus/system procedures for communicating according to an AFC allocation scheme, including station/AP behavior and resource use. This makes it relevant for chipset and device implementation.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

9. Automated frequency coordination and device location awareness

Publication / family	US20220329979A1
Assignee	Arris Enterprises LLC
Inventors	Carol J. Ansley; Charles Cheevers; Ian Wheelock; Thomas Gravely
Status	Pending application
Priority	2019-05-16
Filing	2022-06-28
Publication / grant	2022-10-13
Technical category	Location-aware AFC; AP/gateway geolocation; regulatory response enforcement.

What it did: It tied AFC authorization to device location awareness. The device determines or validates its physical location and uses that information in AFC interaction. This solves a basic AFC requirement: standard-power operation depends on the AFC system knowing where the AP actually is.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

10. Automated frequency coordination for high speed mobility

Publication / family	US20240098615A1 / US12375995B2
Assignee	Qualcomm Incorporated
Inventors	Subramanian Anantharaman; Sreekanth Natarajan; Sandip Homchaudhuri; Sujatha Varadarajan
Status	Published; granted active
Priority	2022-09-15
Filing	2022-09-15
Publication / grant	2024-03-21; grant 2025-07-29
Technical category	Mobility AFC; path learning; mobile power control.

What it did: It adapted AFC for moving/high-speed devices. A device can send AFC requests along a path, use AFC responses to set power, and reduce repeated requests on later trips along the same route. This targets trains, vehicles, robots, or other recurring mobility paths where static AFC assumptions are insufficient.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

11. Automated frequency coordination for high speed mobility

Publication / family	WO2024058874A1 / EP4588288A1 / CN119836828A
Assignee	Qualcomm Incorporated
Inventors	Same Qualcomm mobility inventors
Status	PCT/EP/CN family
Priority	2022-09-15
Filing	PCT 2023-07-31
Publication / grant	WO 2024-03-21
Technical category	International family for high-speed mobility AFC.

What it did: It extends the Qualcomm high-speed mobility AFC invention internationally. The family is important for global connected-transport and industrial-mobility products where the same moving AP/client concepts may be deployed in multiple countries.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

12. Automated frequency coordination (AFC) negotiated band access

Publication / family	US20240171988A1 / US12563405
Assignee	Cisco Technology Inc.
Inventors	John Matthew Swartz; Jerome Henry; Robert Edgar Barton; Matthew Aaron Silverman; Fred Jay Anderson; Vishal Satyendra Desai
Status	Published; granted
Priority	2022-11-21
Filing	2022-11-21
Publication / grant	2024-05-23; issued 2026-02-24
Technical category	Negotiated AFC; future time-window grants; incumbent interaction.

What it did: It changed AFC from pure immediate availability lookup into negotiated future access. A WLAN provider requests spectrum for a future time window; the AFC/coordination system queries an incumbent; if granted, permission is returned to the requester. This supports scheduled, reservation-based, or incumbent-approved spectrum use.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

13. Opportunistic standard power grant by automated frequency coordination

Publication / family	US20240334416A1 / US12439395B2
Assignee	Cisco Technology Inc.
Inventors	Indermeet S. Gandhi; Sudhir K. Jain
Status	Published; granted active
Priority	2023-03-30
Filing	2023-03-30
Publication / grant	2024-10-03; grant 2025-10-07
Technical category	Standard-power grant; low-power to standard-power transition; STA-assisted location.

What it did: It lets an AP start in low-power mode and opportunistically move to standard-power mode when conditions allow. The AP uses a STA/client location token, sends location information to AFC, receives permission, and switches operating mode. This can increase range/performance while staying AFC-compliant.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

14. Opportunistic standard power grant by automated frequency coordination

Publication / family	US20260136350A1
Assignee	Cisco Technology Inc.
Inventors	Continuation family; not fully confirmed
Status	Continuation application
Priority	Continuation of 2023-03-30 family
Filing	2025-10-07 reported
Publication / grant	2026-05-14
Technical category	Continuation strategy for standard-power AFC grants.

What it did: It is a continuation of the opportunistic standard-power grant family. Continuations often pursue additional claim scope around implementation details, such as token handling, STA location use, AP mode switching, or AFC instruction processing.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

15. Automated frequency coordination in wireless communication networks

Publication / family	WO2024153114
Assignee	MediaTek-related PCT filing context
Inventors	Not captured in snippet
Status	PCT publication
Priority	2024-01-17 filing date
Filing	PCT/CN2024/072726
Publication / grant	2024-07-25
Technical category	Controller-based AFC; multi-AP coordination; Wi-Fi network configuration.

What it did: It moved AFC into a multi-AP network-controller architecture. APs or the controller obtain spectrum availability information based on location, and the controller configures multiple APs accordingly. This is relevant to enterprise Wi-Fi and managed networks where AFC should be coordinated across many APs.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

16. Automated Frequency Coordination (AFC) system and method for controlling frequency sharing and/or interference in a communications system

Publication / family	US20240388410A1
Assignee	Charter Communications Operating LLC
Inventors	Abdulrauf Hafeez; Saul A. Torrico
Status	Pending application
Priority	2023-05-16
Filing	2023-05-16
Publication / grant	2024-11-21
Technical category	Threshold-to-interference tables; frequency sharing; interference control.

What it did: It created a precomputed threshold/interference decision approach. The system builds threshold-to-interference tables from transmitter/receiver characteristics before requests arrive, then uses those tables to decide whether a transmitter would cause unacceptable interference. This speeds AFC decisions.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

17. Monitoring of automated frequency coordination network to detect and mitigate successive interference

Publication / family	US20250088294A1
Assignee	Not confirmed from snippet
Inventors	Not confirmed from snippet
Status	US publication
Priority	2023-09-13
Filing	2023-09-13
Publication / grant	2025-03-13
Technical category	AFC network monitoring; KPI-based interference source detection.

What it did: It added ongoing monitoring after AFC grants. The system watches interference indicators and AP KPI data, identifies likely interfering APs, and takes mitigation actions. This is useful because interference problems can appear after initial AFC authorization.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

18. Successive interference mitigation in automated frequency coordination (AFC)

Publication / family	US20240349173A1 / US12598546B2
Assignee	Cisco Technology Inc.
Inventors	Sachin D. Wakudkar; Shailender Potharaju; Evgeny Yankevich; Ardalan Alizadeh
Status	Published; granted active
Priority	2023-04-17
Filing	2023-04-17
Publication / grant	2024-10-17; grant 2026-04-07
Technical category	Successive interference mitigation; AP cause attribution; I/N threshold control.

What it did: It identifies and mitigates aggregate or successive interference from APs. It receives AP channel-use messages, detects when an incumbent fixed-service channel exceeds an I/N threshold, identifies AP contributors, and takes action to reduce the interference.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

19. Destination-based automated frequency coordination

Publication / family	US20250039912A1 / US12550172B2
Assignee	Cisco Technology Inc.
Inventors	Domenico Ficara; Alessandro Ertà; Roberto Muccifora; Amine Choukir; Pascal Thubert; Salvatore Valenza; Arun Khanna
Status	Published; granted active
Priority	2023-07-26
Filing	2023-07-26
Publication / grant	2025-01-30; grant 2026-02-10
Technical category	Client/destination-location AFC; mobile endpoint frequency selection.

What it did: It bases AFC frequency lookup on the destination or mobile client location rather than only the AP location. The system determines the mobile client location, asks AFC for availability at that location, selects a frequency, and causes the client to use it.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

20. Automatic frequency coordination with residential gateway access points

Publication / family	US20250097945A1
Assignee	Charter Communications Operating LLC
Inventors	Deh-Min Richard Wu
Status	Pending application
Priority	2024-11-25 record; related 2022 US noted
Filing	2024-11-25
Publication / grant	2025-03-20
Technical category	Residential gateway AFC; local channel exclusion.

What it did: It embeds AFC/control logic into residential or business gateway APs. The gateway applies local channel exclusion and self-organizes around unavailable unlicensed channels. This targets cable/MSO gateway deployments where AFC needs to work at home/business scale.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

21. Optimizing automated frequency coordination for fixed path uses

Publication / family	US20250247817A1
Assignee	Cisco Technology Inc.
Inventors	Roberto Muccifora; Amine Choukir; Domenico Ficara; Alessandro Erta; Pascal Thubert; Sachin D. Wakudkar; Jerome Henry
Status	Pending application
Priority	2024-01-31
Filing	2024-01-31
Publication / grant	2025-07-31
Technical category	Fixed-path AFC; trajectory prediction; proactive channel selection.

What it did: It optimizes AFC for known fixed routes. The system models path segments/polygons, predicts channel/power availability along the route, and selects channels proactively based on upcoming conditions. It targets rail, guided vehicles, warehouses, mines, and campus transport.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

22. Automatic frequency coordination

Publication / family	US20250274874A1
Assignee	Qualcomm Inc.
Inventors	Parthiban Ellappan; Xiaoxin Zhang; Rishabh Raj; Zhiqun Liu; Tevfik Yucek
Status	Pending application
Priority	2024-02-26
Filing	2024-02-26
Publication / grant	2025-08-28
Technical category	Height/RAT-aware AFC; device-side power/channel control.

What it did: It appears to add context-aware AFC using position/height and radio-access information. The system uses device context to constrain transmit power/channel selection under AFC. This can benefit chipset implementations where height and RAT context affect allowed operation.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

23. Automated Frequency Coordination Registration

Publication / family	US20260040194A1
Assignee	Cisco Technology Inc.
Inventors	Jerome Henry; Peiman Amini; Matthew S. MacPherson; Ardalan Alizadeh
Status	Pending application
Priority	2024-08-05
Filing	2024-08-05
Publication / grant	2026-02-05
Technical category	AFC registration; AP onboarding; fleet deployment.

What it did: It focuses on registering APs or network devices with an AFC system. Device identity, location, and network information are used to onboard equipment efficiently. This is significant for large AP fleets where AFC registration must scale.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

24. Request of automated frequency coordination response

Publication / family	US20260046832A1
Assignee	Hewlett Packard Enterprise Development LP
Inventors	Junpeng Wu; Rajini Balay; Subbiah Nagarajan Veerabudran; Nethra Muniyappa; Zhonglin Feng; Xiumeng Wang; Yiting Zheng
Status	Pending application
Priority	2024-08-09
Filing	2024-08-09
Publication / grant	2026-02-12
Technical category	AFC response request; location samples; AP geolocation evidence.

What it did: It improves how an AP requests an AFC response using location samples, GPS/ranging data, or server-side processing. The goal is to produce a faster or more accurate AFC response based on stronger location evidence.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

25. Automated frequency coordination (AFC) and geolocation enhancements

Publication / family	US20260082363A1
Assignee	Qualcomm Inc.
Inventors	Ahmed Owais Qureshi; Tevfik Yucek; Matthew Tornquist; Hardik Desai
Status	Pending application
Priority	2024-09-16
Filing	2024-09-16
Publication / grant	2026-03-19
Technical category	Vertical uncertainty; multiple height services; geolocation-enhanced AFC.

What it did: It improves AFC by querying multiple height/geolocation services and choosing the smallest vertical uncertainty. Since AFC decisions can depend on AP height and uncertainty, better vertical confidence may increase usable channels or power while maintaining protection.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

26. Automated frequency coordination authentication

Publication / family	WO2025250411
Assignee	Not confirmed from snippet
Inventors	Not confirmed from snippet
Status	PCT publication
Priority	2025-05-20 filing date
Filing	PCT/US2025/030272
Publication / grant	2025-12-04
Technical category	AFC authentication; state-aware operating updates; channel/power enforcement.

What it did: It addresses trust and enforcement around AFC response handling. The device receives available spectrum/power constraints, determines when to update its channel/power based on wireless interface state, and applies the updated operating parameters.

Practical implementation meaning: In a product or AFC service, this type of invention would normally map to one or more modules such as AP geolocation, AFC request generation, server-side incumbent protection calculation, channel/power response enforcement, standard-power mode switching, path/mobility optimization, or monitoring and mitigation after authorization.

Why it matters competitively: It can influence how vendors design Wi-Fi 6E/7 APs, AFC cloud services, WLAN controllers, and chipset firmware for compliant standard-power operation.

Overall Takeaway

Together, these title-matching patents cover the AFC lifecycle: registering the AP, proving location/height, requesting AFC authorization, computing incumbent protection, returning channel/power constraints, enabling standard-power mode, supporting mobile or fixed-path devices, and monitoring/mitigating interference after operation starts.